

Solution Design Document

Early-Warning Financial Distress Screening for a State-Owned Electricity Utility
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The Problem, in Plain Terms

Electricity utilities that are state-owned in many developing nations issue their financial statements on a yearly basis, sometimes with up to one or two years of lag. The moment any red flags appear in the accounts of this organization, there is a possibility that the problem that exists could be at least a year old. This is because the conventional early-warning systems use the history of company failure in order to detect problems, and this is something that cannot be used when dealing with such organizations because there is little history of failure.

What This Solution Does

This project develops a screening model that forecasts financial distress without requiring any history of the utility's failure, based on information that becomes available earlier than the financial reports on an annual basis. The approach works in two ways:

1. Financial distress pattern using a ratio-based risk scorecard developed for other utilities. This method involves learning the financial distress pattern from the group of companies with a similar business profile elsewhere (in this particular case, US utilities with an established history of financial distress) and applying it to the ratios provided by the target company without the need to learn the financial distress of the company itself.
2. Early operational warning indicator well before the release of the annual report. Additionally, it analyzes the amount of power delivered by the company and compares it with the expected amount. Increasing divergence of these two factors can indicate the problems that the company is facing operationally or financially, well before being reflected in the audited financial statement. Both signals were tested against real, independently documented events (a government audit finding for one comparator utility, and real news reporting of a debt crisis and government rescue for the target utility) and both correctly flagged the periods of known distress.

How It Is Built (Solution Architecture)

Data Inputs	Financial ratios of the comparator utilities; Financial ratios of the target utility; Monthly records of electricity deliveries (consumption) of the target utility. All genuine, source-cited data – not artificial or fictional numbers.
Analysis Engine	Two independent computer algorithms: (1) The algorithm of statistical risk assessment trained on the comparator utilities one time and used without changes on the target utility; (2) The forecast model, which learns how the delivery pattern of the target utility looks like normally and reports the growth of deviations from it.
Outputs	Risk score and a handful of plain-language warning thresholds (such as “debt-to-equity greater than X”) that a layman reviewer can test with a calculator; Monthly divergence rate, which shows if something goes wrong operationally. Both are created for regular verification, not a single-time one.
Verification	All figures produced by the tool will be compared with genuinely published external evidence (an auditor's report, press coverage of the debt crisis), not only with internal statistics.

Content of the Code Repository

- **"Data" folder:** The actual input data files (comparator utility ratio, target utility ratio, and target utility monthly deliveries), along with a plain text README file describing the source of each of those files.
- **"notebook.ipynb" file:** The single analysis notebook that recreates all the numbers and figures of the report directly from the data files without any manual editing in between.
- **"Figures" folder:** The figures mentioned in the report (accuracy of risk ranking, feature importance, threshold validation, and delivery divergence trend), created automatically from the analysis notebook. Results summary – a plain-language write-up of what the numbers mean, intended for a reader who will not open the code itself.

Who Would Use This

The development lending institution or the bank can take the risk rating as one of the variables for making their decision on whether or not to lend money. The national energy regulatory agency can apply the four straightforward risk levels without using any software and without having to wait for the next audited balance sheet. The managers at the utility can employ the delivery divergence indicator to differentiate between an internal and an external problem.